

# ECE 3210 Final Exam

*Week of: December 8, 2025*

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You have 3 hours for 6 problems.

- Show enough (neat) work in the clear spaces on this exam to convince us that you derived, not guessed, your answers.
- Put your final answers in the boxes at the bottom of the page.

You may use three US letter-style size page of notes, front and back.

| Problem            | Score | Possible Points |
|--------------------|-------|-----------------|
| 1                  |       | 20              |
| 2                  |       | 20              |
| 3                  |       | 20              |
| 4                  |       | 20              |
| 5                  |       | 20              |
| 6                  |       | 25              |
| <b>Total score</b> |       | 125             |

**1 Short answer**

- (a) Consider a discrete-time LTI system with the transfer functions, difference equations, and impulse responses. Determine if the system is stable, causal, and FIR or IIR.

(i)  $H[z] = \frac{z^3 - z^2 + z - 1}{z^2}$

|             |        |            |
|-------------|--------|------------|
| Circle one: | stable | unstable   |
| Circle one: | causal | non-causal |
| Circle one: | FIR    | IIR        |

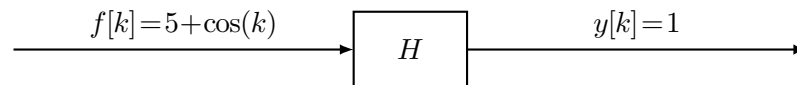
(ii)  $y[k+2] - 1.5y[k+1] + 0.5y[k] = f[k+1]$

|             |        |            |
|-------------|--------|------------|
| Circle one: | stable | unstable   |
| Circle one: | causal | non-causal |
| Circle one: | FIR    | IIR        |

(iii)  $h[k] = (0.8)^k u[k] + (1.2)^k u[k-1]$

|             |        |            |
|-------------|--------|------------|
| Circle one: | stable | unstable   |
| Circle one: | causal | non-causal |
| Circle one: | FIR    | IIR        |

- (b) Consider the input/output relationship of a discrete-time system given by the I/O diagram below. Determine if the system *can* be LTI given the provided information.



Circle one:

LTI

not-LTI

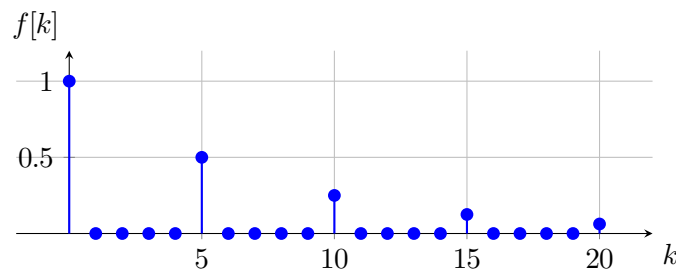
- (c) Find the  $z$ -transform of the signal

$$f[k] = \begin{cases} 0.5^n, & k=5n, \quad n=0,1,2,\dots \\ 0, & \text{otherwise} \end{cases}$$

This signal can also be defined as

$$f[k] = \sum_{n=0}^{\infty} 0.5^n \delta[k-5n].$$

This is shown in the figure below.



$F[z] =$

## 2 Difference equations

Consider the discrete-time system defined by the difference equation

$$y[k+2] + y[k+1] + 0.25y[k] = f[k+1] + f[k]$$

with initial conditions  $y[-1] = -2$ ,  $y[-2] = 4$ , and input  $f[k] = (-1)^k u[k]$ .

- (a) Find the zero-input response  $y_{zi}[k]$ .

$y_{zi}[k] =$

- (b) Find the zero-state response  $y_{zs}[k]$ .

$$y_{zs}[k] =$$

- (c) Find the total response  $y[k]$ .

$$y[k] =$$

### 3 Convolution

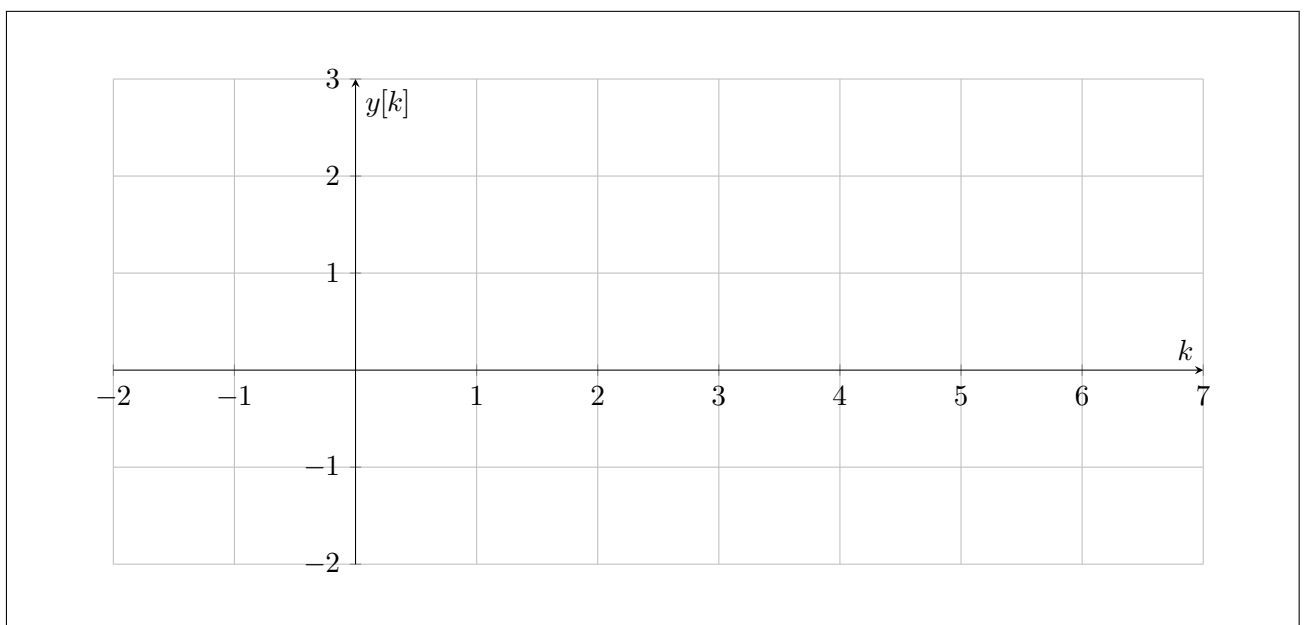
- (a) Derive the convolution

$$y(t) = t \cdot u(t-1) * (u(t) - u(t-2)).$$

$y(t) =$

(b) Derive the convolution and sketch  $y[k]$  in the plot below

$$y[k] = (u[k] - u[k-3]) * \cos(\pi k) u[k-2].$$



#### 4 Sampling

- (a) Suppose you have some signal  $x(t)$  sampled at 50 kHz to produce a discrete-time signal. You then take 1000 samples of this discrete-time signal and compute the 2500-point DFT of these samples. The DFT is a complex-valued array  $F_r$ , where  $r$  is the number of the DFT index (from 0 to 2499). You find that the DFT has a peak at index number  $r=80$ . What is the continuous-time frequency (in kHz) of this peak in the discrete-time signal?

Frequency = \_\_\_\_\_ kHz

- (b) If you have a new signal  $x(t)$  that is bandlimited to 15 kHz, what is the minimum sampling frequency (in kHz) you can use to sample this signal without aliasing?

Minimum sampling frequency = \_\_\_\_\_ kHz

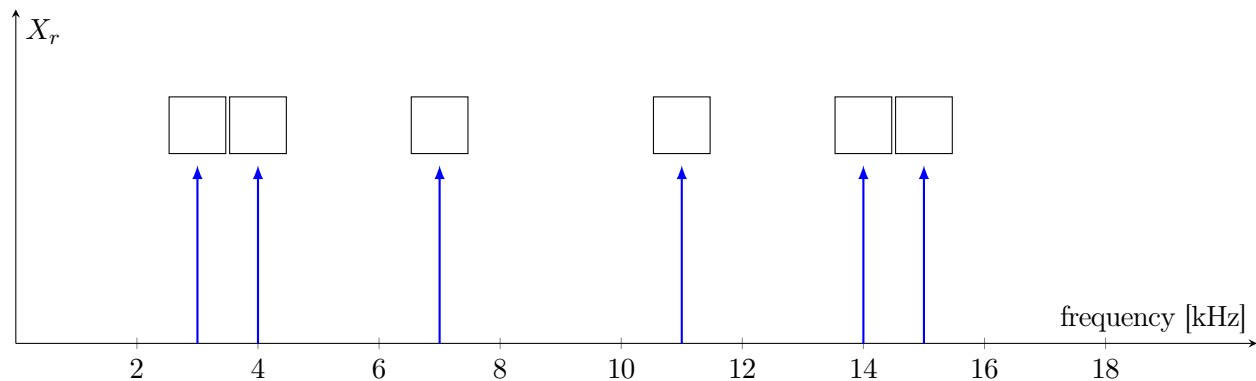


- (c) Suppose you have some continuous-time signal  $f(t)$  that is periodic and built using sinusoids

$$x(t) = \sum_{n=1}^7 \cos(2\pi f_0 n t),$$

where  $f_0 = 11\text{kHz}$ . We sample this signal with a period  $T = 25\mu\text{s}$ . We then reconstruct the sampled signal with an ideal low-pass filter with a cutoff frequency of  $f_c = 16\text{kHz}$  to get a reconstructed signal  $x_r(t)$ .

The plot below shows the Fourier transform  $X_r(\omega)$  of the output signal (though the  $x$ -axis is scaled such that the frequency is in  $f$  kHz rather than  $\omega$  with the relationship being  $f = \frac{\omega}{2\pi}$ ), for frequencies between 0 and 20 kHz. Write the number of the harmonic of  $x(t)$  that produced each component of  $X_r$  in the box above that component. If none of 1-7 could have produced this frequency, enter X. (*Hint*: think about this as if you are in lab lookat the FFT output on the oscilloscope screen after reconstructing your signal.)



**5 DTFT**

(a) Evaluate the summation

$$\sum_{k=-\infty}^{\infty} \text{sinc}^2\left(\frac{\pi}{4}k\right) \cos\left(\frac{\pi}{4}k\right).$$

$$\sum_{k=-\infty}^{\infty} \text{sinc}^2\left(\frac{\pi}{4}k\right) \cos\left(\frac{\pi}{4}k\right) =$$

- (b) Suppose some function  $f[k]$  has a DTFT

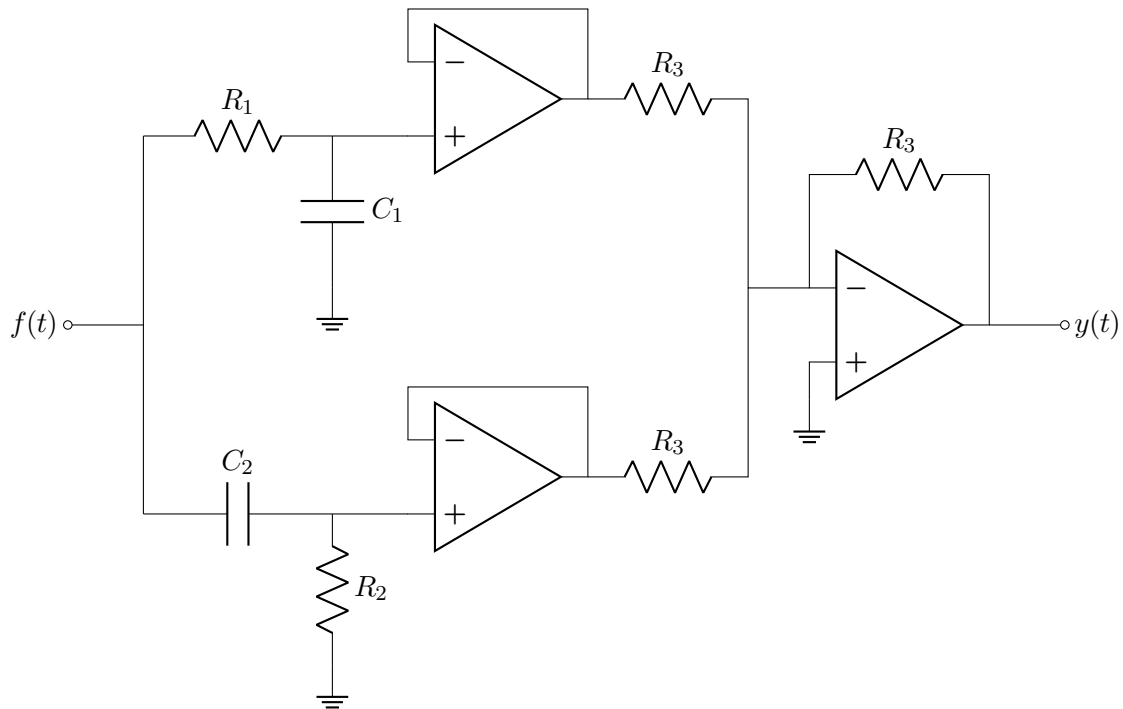
$$F(\Omega) = \frac{\sin(3\Omega)}{\sin(\Omega/2)}.$$

What is  $f[k]$ ?

$f[k]=$

## 6 Continuous-time filters

- (a) Consider the following circuit. Derive the transfer function  $H(s)$  for this circuit in terms of  $R_1$ ,  $R_2$ ,  $R_3$ ,  $C_1$ , and  $C_2$  (if the component is present in the final answer). Simplify your answer as a single ratio of polynomials in  $s$ . (*Hint*: you may find it helpful to analyze this circuit as modular stages.)



$H(s) =$

- (b) Design a low-pass filter using a Butterworth design to hit the following specifications (*Note: this problem has nothing to do with the circuit in the previous part.*):

- Passband edge frequency: 1kHz, minimum passband gain:  $-1\text{dB}$
- Stopband edge frequency: 5kHz, minimum stopband attenuation:  $-20\text{dB}$

Find the minimum filter order  $n$  and the cutoff frequency  $\omega_c$  of the filter.

$n =$

$\omega_c =$

- (c) Using the filter order and cutoff frequency you found in the previous part, find the transfer function  $H(s)$  of the filter. An abbreviated Butterworth polynomial table is provided below.

| $n$ | $B_n(s)$          |
|-----|-------------------|
| 1   | $s+1$             |
| 2   | $s^2+\sqrt{2}s+1$ |
| 3   | $s^3+2s^2+2s+1$   |

$H(s) =$

- (d) If you are designing this as a Sallen-Key filter, you may assume unity gain (i.e., no gain) for each stage, you will have a transfer function

$$H(s) = \frac{1}{s^2 + \frac{R_1+R_2}{R_1R_2C_1}s + \frac{1}{R_1R_2C_1C_2}}.$$

If you are limited to using  $10\text{ k}\Omega$  resistors, what capacitors should you use to implement this filter? (Assume you can magically use any capacitor value you want.)

$C_1 =$

$C_2 =$